

Visal Chathuranga

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Aspiring AI Engineer with hands on experience delivering 20+ AI/ML and full stack projects, including RAG based document intelligence platforms, agentic voice AI systems, healthcare EHR solutions, and production grade SaaS applications. B.Sc. in Electrical & Electronic Engineering from University of Peradeniya, graduated with Second Class Upper Division Honours. Currently developing AI powered healthcare solutions at MedcubeUSA, specializing in end to end ML pipelines, LLM integrations, and intelligent automation across FastAPI, Next.js, and cloud native architectures.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, HTML, CSS, JavaScript

Developer Tools: Git, AWS, Vercel, RStudio

Frameworks: Next.js, Node.js, Flask, Express.js, React, EJS

Libraries: jQuery, TensorFlow, Keras, OpenCV, LangChain, LangGraph, Huggingface, Scikit-learn, MediaPipe, Pandas, NumPy, Matplotlib, Seaborn, NLTK, Prisma, Tesseract.js, OpenAI SDK

Databases & Cloud: MongoDB, PostgreSQL, Pinecone, Neo4j, AWS

MLOps: DVC, MLFlow, CI/CD, Docker, Kubernetes

Additional Skills: OOP Concepts, UI/UX Design, Operations Research, Twilio, WebSocket, REST APIs

EXPERIENCE

Software Engineer - AI/ML

Sep 2025 - Present

MedcubeUSA

- Architected and deployed enterprise grade AI solutions including agentic voice assistants, EHR search tools, and intelligent document processing pipelines. Leveraged RAG, AWS, and Generative AI to automate complex healthcare and financial workflows, enhancing system efficiency.

AI/ML Engineering Intern

Jul 2024 - Oct 2024

GPV Lanka (PVT) Ltd, Sri Lanka

- Built AI-powered desktop apps for reflow oven temperature prediction and feeders management using Flask, ML, and Python.

Automation Engineering Intern

Jul 2023 - Oct 2023

Phoenix Industries Ltd, Makandura, Sri Lanka

- Engineered automated toothbrush packing machine and defect detection system for Astra Cups, improving efficiency and quality control.

PROJECTS

Amazon Textract Invoice PDF Processor (Enhanced with LLM)

[\(Textract Only\)](#) [\(Textract + LLM\)](#)

- Built a scalable invoice automation pipeline using AWS Textract with S3 integration, MD5 duplicate detection, JSON outputs, and intelligent table parsing, achieving 95% accuracy and reducing manual effort by 80%.
- Enhanced the system with a Gemini LLM layer to interpret Textract outputs into Markdown/JSON, normalize dates, currencies, and payment terms, and improve generalization across complex or unseen invoice formats.

MedScribe Pro: Medical Transcription Platform

[\(GitHub\)](#)

- Built a FastAPI platform converting medical audio (.mp3/.webm) into role-labeled transcripts (Doctor/Patient/Nurse) using AWS Transcribe Medical and GPT-4 for clinical summaries.
- Optimized with S3 Transfer Acceleration, multipart uploads, concurrent workers, and speaker diarization for real-time processing.

CRUD Web App

[\(GitHub\)](#) [\(Demo\)](#)

- Built a full-stack MERN application (MongoDB, Express.js, React, Node.js) with secure JWT authentication and RESTful APIs.
- Implemented PDF report generation, dynamic video background, and responsive glassmorphism UI with real-time data updates across all devices.

DocSearch AI: Legal Document Intelligence Platform

[\(GitHub\)](#) [\(Demo\)](#)

- Built an AI-powered legal document search and Q&A platform using a RAG pipeline with OpenAI embeddings and Pinecone vector database, enabling semantic search across case laws, labour codes, and regulations with contextual accuracy.
- Implemented real-time AI chat with WebSocket streaming, OpenAI Assistants API, and per-session vector stores for interactive document-grounded Q&A over uploaded legal PDFs with OCR support (Tesseract.js).
- Developed a full-stack TypeScript/Node.js application with dual databases (PostgreSQL + MongoDB), PDF processing, admin/client dashboards, and role-based access control with JWT authentication.

WhatsApp Petty Cash Management System

[\(GitHub\)](#)

- Built full stack expense system with conversational WhatsApp bot, FastAPI (MVC), and Next.js admin panel featuring AI-powered OCR (AWS Textract + OpenAI GPT) and dynamic rules engine for grade/location based calculations.
- Architected multi tenant SaaS with RBAC, organization provisioning, audit logging, and dual channels (WhatsApp AI + web) for state managed claim workflows with real time notifications.
- Deployed Docker based production system supporting bill based/outright claims with OCR validation, difference warnings, and comprehensive admin panel for approvals and employee management.

EHR AI Search: Healthcare Intelligence Platform

[\(GitHub\)](#) [\(Demo\)](#)

- Built an AI-powered EHR search platform using GPT-4o Mini with natural language queries, React chat widget, and voice I/O support.
- Integrated real EHR API endpoints to search and retrieve patient records from the database with smart fallback for partial matches.
- Deployed as an embeddable Vercel widget with FastAPI backend, WebSocket streaming, and Playwright-based auto-token refresh.

AI Voice Agent with RAG & Telephony

[\(GitHub\)](#)

- Built a production-ready voice AI system integrating OpenAI Realtime API with Twilio telephony for natural phone conversations with real-time audio streaming.
- Implemented a Pinecone-based RAG pipeline with dual modes: Full Agentic (RAG + web search + tools) and RAG-Only for compliance-critical industries.

YouTube Sentiment Insights

[\(GitHub\)](#) [\(Demo\)](#)

- Developed AI-powered Chrome Extension and Flask backend for real-time YouTube comment sentiment analysis, trends, and visuals.
- Built scalable ML pipeline with DVC, MLflow, and LightGBM for experiment tracking, reproducibility, and automated retraining.
- Designed Flask APIs (predict, generate chart, generate wordcloud), integrated into extension, and deployed Dockerized backend on AWS EC2 with GitHub Actions CI/CD.

Sentiment Analysis Pipeline with DVC

[\(GitHub\)](#)

- Designed end-to-end ML pipeline for tweet emotion detection using Python, NLTK, Scikit-learn, achieving 87% accuracy with Gradient Boosting.
- Implemented DVC for tracking datasets, models, metrics; automated stages (Data ingestion → preprocessing → feature engineering → modeling → Evaluating) via dvc.yaml, reducing reruns by 90%.

Industrial AI Solution: Reflow Oven Profile Predictor

[\(GitHub\)](#)

- Designed and implemented a hybrid desktop application using Electron.js frontend with Python Flask backend, integrating TensorFlow/Keras neural networks to optimize soldering temperature profiles for industrial reflow ovens.
- Developed as part of industrial training, this solution demonstrated practical AI implementation in manufacturing processes, resulting in measurable quality improvements.

Industrial Feeders Management System

[\(GitHub\)](#)

- Developed during industrial training, this desktop application provides end-to-end management of industrial feeding systems, combining robust backend functionality with an intuitive operator interface.
- Engineered to optimize production workflows, the system features error-proof UI design and real-time monitoring capabilities for industrial automation environments.

EDUCATION

B.Sc. Engineering (Hons.)

Electrical & Electronic Engineering, University of Peradeniya
Current GPA: 3.4/4.0

2021 – 2025

GCE Advanced Level

Dharmasoka College, Ambalangoda

2010 – 2018

Combined Mathematics: A, Chemistry: A, Physics: B

CERTIFICATIONS

Generative AI Mastery with 15+ Real Time Projects	Udemy
Deep Learning Neural Networks & Image Processing	edXcope
Machine Learning with Python	Coursera (with Honors)
Complete Java Full Stack Training	Digimax Education Sri Lanka
Introduction to Deep Learning & Neural Networks	IBM
Introduction to Computer Vision & Image Processing	IBM
IBM AI Engineering Specialization	IBM
Deep Neural Networks with PyTorch	IBM (with Honors)
Building Deep Learning Models with TensorFlow	IBM
AI Capstone Project with Deep Learning	IBM
Natural Language Processing with Classification and Vector Spaces	DeepLearning.AI
Natural Language Processing with Probabilistic Models	DeepLearning.AI
Introduction to Web Development (HTML/CSS/JS)	IBM
Deep Learning and Reinforcement Learning	IBM

EXTRACURRICULAR ACTIVITIES

Treasurer

IEEE Microwave Theory and Techniques Society (MTTS)

Jan 2024 – August 2025

Programming Team Member

IEEE Microwave Theory and Techniques Society (MTTS)

Jan 2023 – Jan 2024

Member

Explorer's Club, University of Peradeniya

2022 – Present

Volunteer

Actively involved in Arunella, a social development initiative dedicated to supporting school students in underprivileged areas.

2022 – 2025

SOFT SKILLS & INTERESTS

Adaptability	Exploring New Technologies
Problem-Solving	Playing Chess
Teamwork	Environmental Exploration
Communication	Watching Films
Creativity	
Time Management	

NON RELATED REFEREES

Dr. Tharindu Weerakoon

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